Team Builders

Implementing project: LitasCloud (Cloud gaming service)
https://github.com/Team-Builders-PSP/LitasCloud-technical-documentation

Our tech stack:

Player side

- **C#** (.NET 10, Blazor WebAssembly)
- **TypeScript/JavaScript**: WebRTC, Pointer Lock and input module
- **HTML/CSS (Razor)**: UI markdown and styles

Host side

- **C#** (.NET 10, WPF)
- **SQL** (SQLite): local game recordings and settings storage

Server side

- **C#** (.NET 10, ASP.NET Core)
- **SQL**: PostgreSQL
- **YAML**: Docker Compose and GitHub Actions

Requirements we are planning to implement:

- M1 (Host chooses the games he want to offer and a length of the offering)
- M2 (Player can browser games and play them)
- M3 (Host can deny or approve session requests)
- M4 (Player can play a game from browser): we are planning to firstly do a browser client, however if we will run into mouse input streaming problems we will try to do a application client using Electron.
- M5 (Player can use both keyboard and mouse to play a game)
- M6 (Host can end the session whenever they want to)
- M7 (The player will be shown both games FPS, (game to sever) and (client to host) latencies.)
- M8 (Implement registration and login)
- M9 (Host can choose the time of session between 15 minutes and 4 hours. Both player and host shall see the remaining duration of the session)
- M10 (Host can easily add games they want to offer by choosing the game path, the host-side application should verify the games)
- M11 (The game should run hidden from host's screen)
- M12 (Player shall connect directly to host's PC)
- M13 (All game session recordings shall be saved on servers for 7 days unless a report is open so that it can be investigated to look for any form of abuse)
- S1 (Player can choose to cancel his pending session request if the host has not accepted it yet)

- S2 (The application should connect both host and player (open ports, open the connection, close ports after the connection)
- S3 (The host that ends the connection early shall be punished and cannot start another session for an hour, if such infraction repeats over the course of the next 30 days host shall be prevented from opening another session for 24 hours, unless a justification for such actions is presented)
- S4 (The price for the session only should be calculated by the played time and the player should be charged fairly and a host should be paid full amount minus 5% fee)
- S5 (Host's PC shall be tested for its' legibility to stream gaming sessions before lease starts. Internet connection speed, hardware and software shall be tested)
- S6 (The service should not disconnect the both ends if there is a short connection drop)
- C1 (Game sound)
- C5 (System shows host's PC's specifications before joining the session)
- C6 (Player can ask for more session time, host can accept or decline within 30 seconds, total session time maximum limit still remains 4 hours)
- C7 (Rating and reports after session for the admins)
- C9 (Both player and host can see their session history, but cannot see recordings of these sessions)

Total of **24** requirements we are planning to implement